

Satellite Inference

Mission-assured inference where space data begins

Public Whitepaper v0.3



Notional concept render. This is not flight CAD and does not establish dimensions, interfaces or launch compatibility.

PUBLIC CONCEPT

The first owned mission target is a 10 kW continuous compute-payload node in low Earth orbit. The product thesis is to convert live space-originated data into compact, decision-ready outputs before ordinary ground processing can respond.

Prepared by Satellite Inference

Satellite Inference is currently operated by RFID INC, a Delaware corporation.

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READER GUIDE

1. What this paper does

This paper explains the company thesis, the first customer problem, the 10 kW LEO concept, and the evidence path from a ground tile to a flight mission. It is deliberately shorter than the technical appendix. Detailed design tables, interface allocations and verification matrices belong in the Mission Definition, not here.

Evidence label	Meaning
PROGRAM TARGET	A capability the program intends to demonstrate.
PUBLIC EVIDENCE	A statement supported by an official or openly available source.
WORKING ASSUMPTION	A planning value that still requires analysis, supplier data or test.
UNKNOWN	A decision-critical item that is not yet established.

The honest boundary

- No flight spacecraft, launch booking or customer service commitment is represented as complete.
- The 10 kW level is a system target, not evidence that one detector needs 10 kW.
- The first flight requires a lawful, technically closed route from a live or operationally representative sensor source.
- Ground demonstrations reduce risk, but do not replace environmental qualification or on-orbit evidence.

TECHNICAL APPENDIX

For technical parameters and verification intent, read the separate 10 kW Orbital Edge Node Mission Definition v0.2 [8].

WHY COMPUTE IN ORBIT

2. The thesis

10 kW	500-600 km	1 -> 10 kW
AGGREGATE COMPUTE INPUT	LEO TRADE SPACE	GROUND EVIDENCE PATH

Most model training, archive processing and non-urgent analytics belong on Earth. Orbital inference earns a role only when proximity to the sensor changes a measurable outcome: fewer transmitted bits, a shorter observation-to-decision interval, autonomous prioritization, or assured processing when connectivity is constrained.

Satellite Inference proposes a shared, fault-tolerant compute node that ingests authenticated space-originated data, executes versioned customer-approved workloads, and returns compact products with provenance. Those products may be a vessel track, an event alert, a selected image chip, a quality score or a request for another observation.

What the company would sell

- Reserved processing capacity for a defined mission workflow.
- Response-time, availability and accepted-output commitments.
- Data custody, workload control, provenance and service reporting.
- Integration with a customer or partner sensor data path.

PRODUCT POSITION

The commercial unit is an accepted mission outcome, not a commodity GPU-hour. Generic cloud inference is a later option, not the first wedge.

THE PRODUCT LOOP

3. From sensor to decision

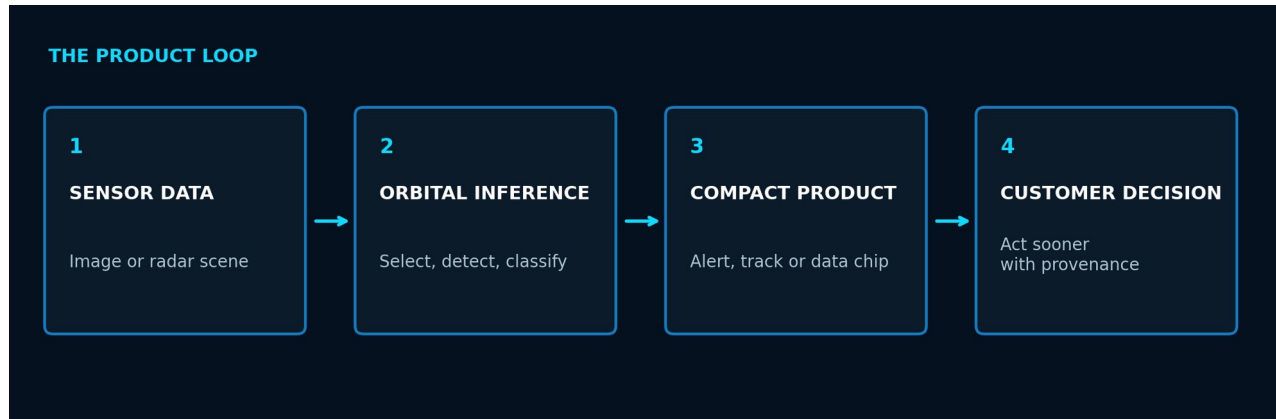


Figure 1. The proposed service converts a selected space observation into a smaller, decision-ready product.

Why the loop matters

- A sensor creates the original observation. Satellite Inference does not need to own that sensor for the shared-compute model.
- Inference means running an already-trained model. Training large models is not required for the first product.
- Downlink carries data from spacecraft to Earth. A crosslink carries data between spacecraft.
- The node creates value only if the result reaches an operational workflow in time to change a decision.

NASA's FAME project describes onboard edge computing that analyzes imagery in real time and coordinates follow-up observations across multiple sensors [7]. ESA's PhiSat-2 applications demonstrate onboard vessel detection, wildfire detection and cloud screening [4][5][6]. These are technical precedents, not proof of demand for an independent 10 kW service.

FIRST BENCHMARK CANDIDATE

4. SAR maritime vessel detection and scene prioritization

Synthetic aperture radar, or SAR, can image vessels through clouds and darkness. The primary benchmark candidate is to turn an analysis-ready SAR scene into geolocated vessel detections, confidence scores, approximate size where supported and a scene priority, then deliver only the result and selected review chips rather than the entire scene.

A single SAR scene can support vessel presence, location, approximate size and scene prioritization. It cannot by itself establish identity, intent or illegal activity. Activity cueing requires temporal observations and permitted context such as AIS, RF data or analyst review.

Input	On-node work	Decision product
Analysis-ready SAR scene plus permitted context	Tiling, normalization, vessel detection, classification, confidence scoring and optional track correlation	Geolocated detections, confidence, selected chips, provenance and audit sample

Why this scenario leads

- The customer problem is specific: detect vessels across agreed environmental and size regimes, then prioritize scenes for human or downstream review.
- The workload has a public evaluation path. xView3 provides a large labeled SAR benchmark for vessel detection and characterization [3].
- Operational relevance is visible. DIU reports that xView3 algorithms were integrated into maritime-domain awareness platforms [2].
- Public procurement is adjacent evidence. In August 2026, the NRO announced three commercial radar augmentation contracts for intelligence, monitoring, disaster response and humanitarian requirements [1].

COMMERCIAL UNKNOWN

No public source proves that a customer will buy independent 10 kW orbital compute. The benchmark becomes a mission only after data rights, a viable ingress route, acceptance metrics and a paying design partner are closed.

REUSE THE PLATFORM, DO NOT CREATE THREE COMPANIES

5. Two secondary scenarios

Scenario	Customer value	Role in v0.3
Wildfire and rapid-change detection	Create an early alert, event footprint or follow-up request before the full image reaches a ground archive.	Secondary benchmark for time-critical event detection and cross-cueing.
Optical image quality and cloud triage	Discard unusable observations, rank clear images and pass only selected products to later workloads.	Control benchmark for efficiency, scheduling and pipeline composition.

ESA describes PhiSat-2 applications that detect wildfires and reduce unnecessary downlink, and a cloud-screening service that can feed other onboard applications [5][6]. NASA FAME describes rapid event detection and coordinated follow-up observations [7]. These precedents support the workflow logic.

Selection rule

1. Run all three workloads on the ground with representative data and one measurement protocol.
2. Measure accepted quality, response time, throughput, energy per scene, data reduction and recovery behavior.
3. Select one flight workload only after a customer accepts the benchmark and the sensor route is credible.

PORTFOLIO DECISION

SAR vessel detection and scene prioritization lead the public story. Wildfire/change and optical triage remain reusable workloads, not equal product lines.

A MATERIAL SHARED NODE

6. The 10 kW LEO concept

PROGRAM TARGET: the first owned spacecraft provides 10.0 kW of continuous electrical input at the compute-payload DC interface during nominal operation, including the approved eclipse case, in a 500-600 km low Earth orbit trade space.

Ten kilowatts is aggregate system capacity. It includes accelerators, host processors, memory, storage, networking, local control and compute-side conversion losses. It does not mean 10 kW of GPU silicon, and it does not assert that one vessel detector consumes the full node continuously.

What aggregate capacity can support

- Parallel scenes or customer queues instead of one serial demonstration.
- Preprocessing, inference, fusion, packaging and audit workloads in the same service window.
- Redundancy, fault isolation, checkpoints and graceful throttling.
- Burst demand across maritime, event detection and image-quality pipelines.

Why LEO first

LEO matches the first Earth-observation data sources, supports lower communications latency than distant orbits, and keeps the mission focused on a near-term sensor-to-decision service. GEO and Lagrange points are not steps in the v0.3 first-mission plan.

ENGINEERING REALITY

At this power level, heat rejection, power generation and storage, radiation tolerance, data movement, deployment dynamics and launch integration are first-order spacecraft problems. The Mission Definition [8] holds the current screening assumptions and verification intent.

BUILD PROOF BEFORE FLIGHT

7. Evidence path: 1 kW to 10 kW

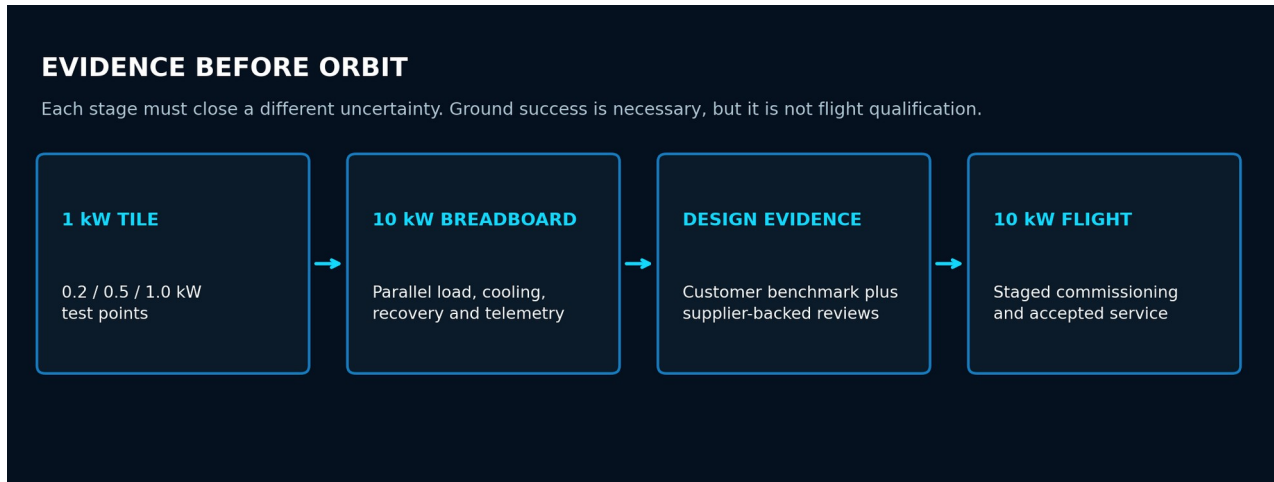


Figure 2. The ground program isolates component behavior, then proves system-scale operation before flight commitment.

The 1 kW engineering tile is a laboratory module, not a smaller spacecraft product. Its 0.2, 0.5 and 1.0 kW operating points reveal efficiency, thermal response, throttling and recovery behavior. The next ground article combines ten tiles, or a functionally equivalent architecture selected after test, into a continuous 10 kW breadboard.

What each stage must prove

Stage	Evidence produced
1 kW tile	Measured compute, power, cooling, networking, telemetry and fault recovery across controlled operating points.
10 kW breadboard	Sustained parallel workloads, full-load heat removal, orchestration, security controls and customer benchmark results.
Design reviews	A coherent spacecraft concept supported by analysis, supplier information, customer requirements and verification planning.
Flight	Staged commissioning, continuous node operation, live sensor-to-result service and safe end-of-life behavior.

A WORKLOAD IS NOT A PRODUCT UNTIL IT ENTERS A WORKFLOW

8. Customer proof before flight

A useful first mission starts with a customer problem, not a processor. The company should first obtain representative data, define a trusted ground reference, measure the same workload on the tile and breadboard, and agree which outputs are acceptable. Only then can a reservation or letter of intent describe a credible future service.

Minimum pre-flight evidence

- A paid ground evaluation or equivalent customer-funded scope.
- Rights to use representative data and an agreed acceptance protocol.
- Measured quality, response time, throughput, energy, data reduction and fault recovery.
- A technically credible same-platform interface or inter-satellite data route.
- A conditional commercial commitment tied to commissioning and accepted performance.

What an early demonstration should show

For the SAR vessel-detection scenario, the demonstration should accept a representative scene, produce vessel detections with confidence and provenance, assign a scene priority, select audit chips, and compare the compact output against the original data volume and ground reference. The result must be reproducible. A video alone is not evidence.

DO NOT OVERCLAIM

A prerecorded upload can prove compute and software behavior. It cannot prove commercial data access, operational latency or a live sensor service.

OWN THE MISSION OUTCOME

9. Operating model and first flight

Satellite Inference should own system requirements, the compute runtime, workload control, customer acceptance, data policy and mission authority. Specialized partners can initially provide the spacecraft bus, power and thermal hardware, communications, environmental test, launch integration and routine flight-dynamics or ground-station services.

Staged commissioning

1. Establish positive spacecraft control, communications, energy margin and thermal telemetry.
2. Activate one compute tile and verify workload execution and recovery.
3. Increase load in controlled steps while preserving automatic limits and safe states.
4. Operate the aggregate 10 kW payload through the agreed nominal case.
5. Run the customer-approved sensor-to-result service and obtain acceptance evidence.

Mission success

Success is not merely powering a processor. It is controlled spacecraft operation, repeatable customer-accepted output, fault containment, auditable data handling, service reporting and retained end-of-life capability. Detailed mission objectives and verification ownership remain in the technical appendix [8].

REPLICATE EVIDENCE, THEN POWER

10. Scaling strategy

Scale step	What changes	Required proof first
First 10 kW node	One material shared service in LEO	Accepted workload, data route, system evidence and safe operations
Fleet of repeatable 10 kW nodes	Coverage, resilience, customer isolation and aggregate capacity	Stable operations, repeatable integration and contracted utilization
100 kW-class architecture	Larger power and thermal system or clustered modules	Demand that cannot be met economically by replicated 10 kW nodes
Network-scale capacity	Multi-orbit routing, automated scheduling and standardized production	Demonstrated service economics and mature manufacturing

The governing rule

Scale follows contracted workflows and data routes. A larger spacecraft without paid utilization only concentrates technical and schedule risk. The most credible path is to standardize the 1 kW tile, prove one 10 kW node, replicate where coverage or resilience justifies it, and move to 100 kW only when the workload mix and operations data require it.

SCALING THESIS

The long-term product is an orbital inference network. The near-term job is to prove one trusted sensor-to-decision loop on a repeatable node.

WHAT V0.3 DOES NOT PRETEND TO KNOW

11. Risks, unknowns and next evidence

Open issue	Why it matters	Evidence that closes it
Live data ingress	A free-flying node without sensor access is idle infrastructure.	Partner data rights, a route design and an end-to-end demonstration.
Thermal rejection	Nearly all compute power becomes heat that must reach radiating surfaces.	Correlated ground tests and an orbital thermal design.
Radiation and reliability	High-performance electronics must survive the mission environment and recover from faults.	Environment definition, parts strategy, test and fault-injection evidence.
Power and deployment	Continuous compute through eclipse drives arrays, storage, structure and attitude control.	Coupled analysis, supplier-backed architecture and deployment verification.
Launch integration	Mass, stowed geometry and structural loads determine actual compatibility.	Controlled geometry, mass properties and provider review.
Demand and utilization	No public award establishes demand for this exact service or power level.	Paid evaluations, accepted benchmarks and conditional commitments.

Immediate decision sequence

1. Complete the SAR vessel-detection benchmark specification and run the first controlled 1 kW tile tests.
2. Secure one data-path design partner and one buyer-side design partner.
3. Translate measured workloads into compute, communications, power and thermal requirements.
4. Use those requirements to close supplier studies and design reviews before any binding launch commitment.

SOURCES

12. Public evidence and technical appendix

The sources below support the customer problem, technical precedents and current public concept boundary. They do not establish a customer commitment, supplier selection, qualified spacecraft design or commercial price for Satellite Inference.

- [1] National Reconnaissance Office, Radar Commercial Augmentation contract announcement, 5 August 2026. [Open source](#)
- [2] Defense Innovation Unit, operational use of xView3 maritime algorithms, 3 August 2023. [Open source](#)
- [3] xView3 public SAR vessel-detection benchmark. [Open source](#)
- [4] European Space Agency, AI for marine vessel detection. [Open source](#)
- [5] European Space Agency, AI for wildfire detection. [Open source](#)
- [6] European Space Agency, AI for cloud detection. [Open source](#)
- [7] NASA Earth Science and Technology Office, FAME dynamic targeting project. [Open source](#)
- [8] Satellite Inference, 10 kW Orbital Edge Node Mission Definition v0.2. [Open source](#)

Read next

Technical appendix: [10 kW Orbital Edge Node Mission Definition v0.2](#)

Interactive concept and publications: satelliteinference.com

IMPORTANT

This public paper is a concept and evidence plan. It is not flight-release documentation, an interface specification, a supplier commitment, a launch reservation or legal advice. Technical values remain subject to analysis, test, supplier data, customer requirements and applicable approvals.